

Student Capstone Project

ML, Deep Learning & NLP Applications

Tuwaiq Academy — Bootcamp Programme

Course	Machine Learning, Deep Learning & NLP Applications
Level	Beginner–Intermediate (Grade 12)
Weight	40% of Final Grade
Deadline	End of Week 10 — submitted via LMS
Format	Jupyter Notebook + Written Report (PDF or Markdown)
Team Size	Individual — your own work only

In this project you will build a complete machine-learning pipeline — from raw data to a trained model — and write a short report explaining what you did and what you found. You will use skills from all 10 weeks of the course.

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Section 1 — Overview & Learning Goals

By the end of this project you will be able to:

- ✓ Load and explore a real-world dataset
- ✓ Clean and prepare data for machine learning
- ✓ Train and compare at least 3 different ML models
- ✓ Build and train a neural network
- ✓ Evaluate models using accuracy, precision, recall, F1-score, and confusion matrices
- ✓ Understand and explain your results in plain language
- ✓ Apply concepts from all 10 weeks of the course

Section 2 — Pick Your Dataset

Choose **one** of the three datasets below. Pick the one that interests you most.

Dataset	What it contains	Your goal	Type
Titanic (Kaggle)	Passenger info: age, sex, class, fare — ~890 rows	Predict who survived (1) or not (0)	Tabular
IMDb Movie Reviews	50,000 movie reviews labelled positive or negative	Classify review sentiment	Text
Fake News Detection	News headlines + body text labelled real or fake	Detect whether a news article is real or fake	Text

Section 3 — Your Three Steps

Follow the three steps below in order. Each step builds on the one before it.

Step	What you do	Weeks	Points
1	Machine Learning Models	1–4	35
2	Neural Network	5–8	35
3	Written Report	9–10	20 + 10*

* 10 pts for code quality

Step 1: Machine Learning Models

Weeks 1–4 | 35 pts

In this step you will load your dataset, explore it, prepare it, train at least three different models, and compare their results.

Task 1.1 — Load and Explore

Do the following:

- Load the dataset with pandas and print the first few rows
- Check the shape (number of rows and columns)
- Check for missing values — fill or drop them
- Check class balance: are there roughly equal numbers of each label?
- If one class has far more samples than the other, note this and use F1-score as your main metric

Task 1.2 — Prepare the Data

Do the following:

- Separate features (X) and label (y)
- Split into train (80%) and test (20%) sets — use `random_state=42`

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

Text dataset only (IMDb / Fake News): If you chose IMDb or Fake News, you must also clean the text and convert it to numbers before training. Do the following extra steps:

```
# --- Text datasets only: clean the text ---
import re, nltk
from nltk.corpus import stopwords
nltk.download('stopwords')
stop_words = set(stopwords.words('english'))

def clean(text):
    text = text.lower()
    text = re.sub(r'^a-z\s', '', text) # remove punctuation
    words = [w for w in text.split() if w not in stop_words]
    return ' '.join(words)

df['clean_text'] = df['your_text_column'].apply(clean)

# --- Convert text to numbers with TF-IDF ---
from sklearn.feature_extraction.text import TfidfVectorizer

tfidf = TfidfVectorizer(max_features=5000)
X_train = tfidf.fit_transform(X_train) # learn vocab on train only
X_test = tfidf.transform(X_test)      # apply to test (no fitting!)
```

Remember: Always use `fit_transform()` on the training set and `transform()` on the test set. Never fit the TF-IDF on test data — that is called data leakage.

Task 1.3 — Train and Compare Models

Do the following:

- Train at least 3 models: e.g. Logistic Regression, Random Forest, KNN
- For each model report: accuracy, precision, recall, F1-score
- Plot a confusion matrix for each model
- Put all results in one summary table and choose the best model — explain your choice

Tip: Put all model results in one summary table so they are easy to compare side by side.

Step 2: Neural Network

Weeks 5–8 | 35 pts

Build and train a neural network on the same dataset you used in Step 1. Use the same `X_train` and `X_test` — no new data preparation is needed.

Task 2.1 — Build and Train

Do the following:

- Build a neural network using Keras Dense layers with ReLU and Sigmoid/Softmax
- Add at least one Dropout layer to reduce overfitting
- Train for at least 10 epochs
- Plot the training vs. validation loss and accuracy curves

Remember: If training accuracy keeps going up but validation accuracy goes down, your model is overfitting. Try increasing Dropout or reducing layers.

Task 2.2 — Compare with Step 1

Do the following:

- Evaluate the neural network on the test set using the same metrics as Step 1
- Add the neural network results to your summary table from Step 1
- Write 2–3 sentences explaining whether the neural network was better or worse and why

Tip: Try one improvement: add more layers, change the learning rate, or add more Dropout — then compare before and after.

Step 3: Written Report

Weeks 9–10 | 20 pts

Write a short report (1–2 pages, PDF or Markdown) that summarises your project. Your report must cover all six sections below. **Do not paste code into the report** — explain everything in plain language.

1. Introduction

Example: I chose the IMDb dataset because I wanted to learn how computers can detect opinions in text.

2. Dataset Description

Example: The dataset has 50,000 rows: 'review' (text) and 'sentiment' (positive/negative). Classes are balanced — 25,000 of each.

3. Models Used

Example: I trained three models: Logistic Regression ($F1 = 0.88$), Random Forest ($F1 = 0.86$), and KNN ($F1 = 0.79$). Logistic Regression was best.

4. Neural Network

Example: My network had two Dense layers (128 and 64 neurons) with Dropout. After 15 epochs validation accuracy reached 89%, slightly above Logistic Regression.

5. Results & Comparison

Example: The neural network had the highest F1-score (0.91) but took much longer to train. For this size of dataset the extra time is worth it.

6. What I Learned

Example: More complex models are not always better. Cleaning the text before training made the biggest difference to accuracy.

Section 4 — Grading Rubric

Area	Pts	Full marks	Partial	Zero
Data Loading & Exploration	10	Shape, dtypes, missing values checked; class balance noted	Loaded but minimal exploration	Not done
Data Preparation	10	Split correct; text cleaned & TF-IDF applied (text datasets)	Split done; text prep missing/incorrect	Not done
Step 1 — ML Models	25	3+ models; all metrics reported; confusion matrix; best model justified	2 models; some metrics missing	0–1 model; no metrics
Step 2 — Neural Network	25	Built with Dropout; 10+ epochs; curves plotted; compared with Step 1	Trained but curves or comparison missing	Not attempted
Step 3 — Written Report	20	All 6 sections present; plain language; clear conclusion	4–5 sections; conclusions vague	Missing or very incomplete
Code Quality	10	Notebook runs top-to-bottom; functions used; markdown explains each step	Runs with minor fixes; limited explanations	Does not run
TOTAL	100			

Section 5 — What to Submit

Put everything in a folder called **capstone_yourname/** and zip it:

File	What it contains
notebook.ipynb	Your Jupyter notebook — must run top-to-bottom without errors
report.pdf or report.md	Your written report covering all 6 sections
data/ (folder)	Your raw dataset file (CSV etc.)
requirements.txt	List of every Python package you imported

Tip: Use clear headings in your notebook like '## Step 1 — Logistic Regression' so the marker can follow your work easily.

Section 6 — Academic Honesty

What you **MUST** do:

- Write and run all code yourself
- Cite any code you adapted from tutorials (add a comment with the URL)
- You may use AI tools to understand concepts, but write your own code and report

What is **NOT** allowed:

- Copying another student's notebook or report
- Submitting AI-generated code or text as your own without understanding it
- Sharing your notebook with classmates before the deadline

Section 7 — Glossary

Key terms used in this project:

Accuracy	The percentage of predictions your model got right.
Class Imbalance	When one label appears much more often than the other. Use F1-score as your main metric in this case.
Confusion Matrix	A table showing true positives, true negatives, false positives, and false negatives.
Data Leakage	Accidentally using test data information during training, making results unreliable.
Dropout	Randomly turns off neurons during training to prevent overfitting.
Epoch	One full pass through the entire training dataset during neural network training.
F1-score	The harmonic mean of precision and recall. Best metric when classes are imbalanced.
Overfitting	When a model performs well on training data but poorly on new, unseen data.
Precision	Of all the times the model said 'yes', how often was it correct?
Random State	A seed number that makes random operations repeatable.
Recall	Of all the actual 'yes' cases, how many did the model find?
Stratify	Splitting data so each split has the same class proportions as the full dataset.
TF-IDF	Term Frequency–Inverse Document Frequency: converts text into numbers based on word importance.
Transfer Learning	Starting with a model already trained on a large dataset and adapting it for your specific task.
Validation Set	A portion of training data held back to monitor training and detect overfitting.